

Guideline

The treatment of status epilepticus in the neurosciences critical care unit (NCCU)

1 Scope

Patients admitted to the neurosciences critical care unit with status epilepticus.

2 Purpose

To describe the management of patients with status epilepticus.

3 Definitions

3.1 Electro-encephalography (EEG)

EEG is a recording of electrical activity along the scalp, measuring voltage fluctuations resulting from ionic current flows within the neurons of the brain, usually looking for epileptiform activity.

3.2 Generalised convulsive status epilepticus

This is convulsions with generalised rhythmic jerking of the limbs and mental status impairment, which are prolonged or self-sustaining without recovery of consciousness between seizures.

3.3 Non-convulsive status epilepticus:

This is seizure activity on EEG without clear convulsive clinical features; this includes severely ill patients with 'subtle status' – note that 'wandering confused' patients are not included here.

3.4 Status epilepticus (SE):

This is defined by seizures of a duration > five minutes according to revised definition³. This will inevitably include some patients with prolonged seizures not fulfilling traditional criteria i.e >30minutes duration⁴.

3.5 Established SE:

SE is initially treated with benzodiazepines. Unfortunately, approximately 35-45% of patients are refractory to benzodiazepines. Benzodiazepine-refractory SE is also called established SE¹.

3.6 Refractory SE

Status epilepticus that does not respond to traditional or first-line treatment regimens that includes an adequate trial of one benzodiazepine and one appropriate non-benzodiazepine anticonvulsant. Convention of diagnosis is status epilepticus exceeding two hours despite treatment^{1,2}. Refractory SE is treated by admission to the ITU for intubation and seizure suppression with an intravenous anaesthetic.

3.7 Super-refractory SE

SE which recurs or continues despite adequate continuous infusion of an intravenous anaesthetic for more than 24 hours. True super-refractory SE most often occurs with underlying neurological disorders (commonly: autoimmune encephalitis, a likely cause of NORSE = new onset refractory status epilepticus).

A putative mechanism is that GABA receptor activity decreases and the number of glutaminergic receptors increases (theoretical implications for treatment).

4 Introduction

Over half of patients who experience status epilepticus do not have epilepsy. The incident rate is 10-60 per 100,000 population with higher instances in deprived populations; uncontrolled seizures and status epilepticus constitute 2.2% of intensive care admissions in the UK³. In children, the major cause is infection. In adults, the main causes are:

- Stroke
- Hypoxia
- Metabolic derangements
- Alcohol intoxication or withdrawal (the most common cause in young adults)

It is important to note that patients with a diagnosis of epilepsy may have an acute cause for their seizures.

5 Undertaken by (staff groups)

Medical and non-medical prescribers (within scope of practice) on the NCCU.

6 Criteria for intensive care admission

a) Status epilepticus that fails to terminate after emergent initial therapy: Pharmacological treatment:

- Adequate dose of one of the following benzodiazepine treatments:
 - Lorazepam - 4mg IV (reduce to 2mg in debilitated patients)
 - Midazolam - 10mg buccally, IV or IM
 - Diazepam - 10mg/ 0.15-0.25mg/kg (usually 10mg) over a minute intravenous emulsion (Diazemuls®) or rectal

- Adequate loading dose of non-benzodiazepine anticonvulsant
 - **40mg/kg** of IV levetiracetam over 10 minutes (maximum 3000mg; maximum 4500mg in the first 12 hours – loading and first maintenance dose). Note: there is as yet limited evidence for the use of higher loading doses¹.
 - **40mg/kg** of IV sodium valproate over 10 minutes
 - **20mg/kg** of IV phenytoin – max 50mg/min (maximum 2000mg; round down to the nearest 250mg)
 - See also [phenytoin in adults guideline](#)
 - Empirical 50ml of glucose 50% intravenously
 - Empirical intravenous vitamins: Pabrinex one pair of ampoules IV if malnourished or alcohol-related
- b) Conscious level that remains depressed and ongoing epileptiform activity present on EEG.
- c) Respiratory depression following benzodiazepine therapy that cannot be safely managed in the clinical area. It is likely that patients who have received 3mg/kg or 30mg of diazepam or equivalent in less than 24 hours will require prolonged ventilatory support.

Note: The most common causes of pseudo-refractory SE are:

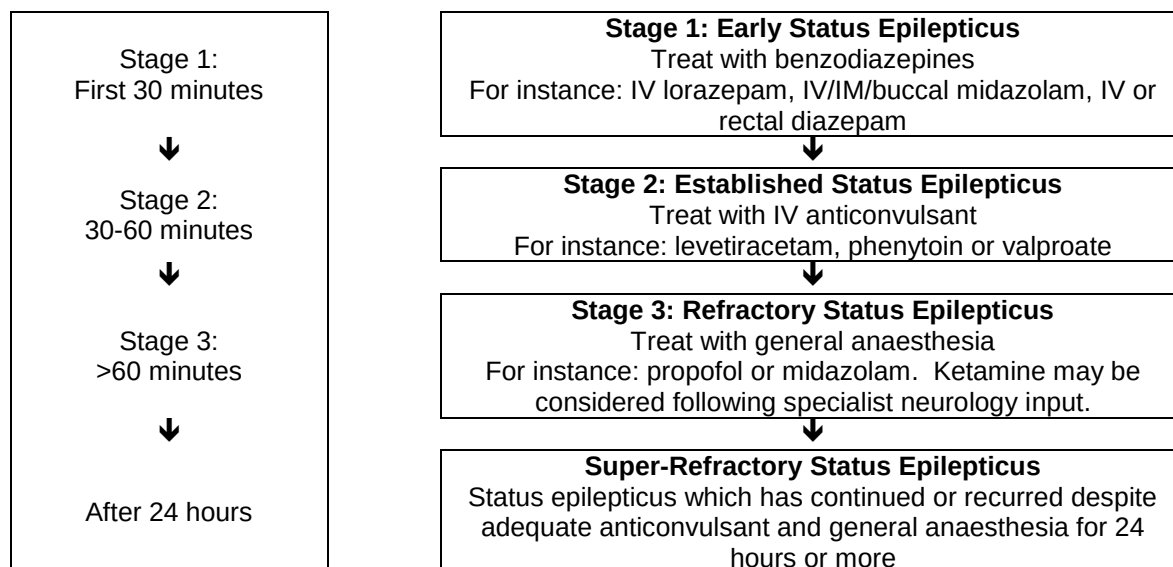
1. **Inadequate drug dosage:** failure to suppress electroencephalographic seizure activity
2. **Wrong diagnosis:** consider non-epileptic 'attacks' or paroxysmal autonomic dysregulation.

Following loading with an IV anticonvulsant, the first maintenance dose can be commenced:

- | | |
|-----------------|----------|
| • Levetiracetam | ≥4 hours |
| • Phenytoin | ≥6 hours |
| • Valproate | ≥6 hours |

¹ The ESETT trial used 60mg/kg, but found mortality differences and did not describe transition to maintenance dosing. N Engl J Med 381; 22: 2103-2113.

7 Management



7.1 Stage 1: Initial management (first 30 minutes)

- Position patient to avoid injury
- Respiratory support with oxygen
- Establish intravenous access
- Replace fluids

7.1.1 Order

- U&E – sodium, potassium, calcium, magnesium and glucose
- FBC – platelets and clotting screen
- Arterial blood gas
- Chest radiograph – to evaluate the possibility of aspiration
- Toxicology – if cause of SE is unknown
- Request serum drug levels for any of the following anticonvulsants* if clinically indicated (test for non-compliance):
 - Carbamazepine
 - Phenytoin (total)
 - Valproate (valproic acid)
 - Phenobarbital
- Levels for other antiepileptic medicines are undertaken by an external laboratory and may be undertaken if there is a particular need:
 - Lamotrigine
 - Levetiracetam
 - Lacosamide
- 5ml serum and 50ml of urine for potential future analysis

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- AED tests unavailable at CUH can be sent to the National Drug Monitoring Laboratory, Chalfont Epilepsy Centre. Liaise with biochemistry/ pathology helpdesk.

7.1.2 Continuous EEG monitoring

This is mandatory for patients with depressed conscious level due to the need to evaluate for cerebral electrical seizure activity where convulsions cease (either clinically overt and/ or EEG proven).

7.1.3 Standard ICU monitoring:

According to standard practice.

7.2 Stage 2: Subsequent management (beginning in 60 minutes)

1. Repeat bloods to exclude metabolic disorders
2. Optimise initial anticonvulsant therapy (levetiracetam / phenytoin/ valproate). If an inadequate dose has been given, the remainder should now be given.
3. Continue regular anticonvulsants (if taken prior to admission).
4. Begin continuous EEG monitoring: to avoid misinterpreting non-epileptic jerking
5. Begin a continuous infusion of general anaesthetic (one of the treatments described in table 1):
 - a. Treatment goal: definite seizure suppression or burst suppression
 - b. The BIS monitor may show one or two channels of raw EEG data – be aware, BIS is of low quality and will miss focal seizure activity which should be suppressed completely to avoid recurrence of SE. Artifact recognition in BIS output is also difficult. Conventional EEG is preferable.
6. Conduct continuous EEG monitoring for at least 24 hours. Continuous EEG monitoring or frequent intermittent EEG is mandatory. EEG findings should be discussed with neurologist at bedside or remote viewing.
7. If pyrexia, consider antipyrexial pharmacotherapy (e.g. paracetamol) or other measures to reduce patient temperature, and commence anti-infective therapy as appropriate.

7.3 Cautions

1. **Avoid neuromuscular blockers** to allow for subtle clinical seizure activity. If required, **always** have EEG monitoring. Be aware of the limitations of BIS. Reduced montage (8-channel)-EEG is sufficient.
2. **Propofol doses** above 5mg/kg/hr are strongly associated with propofol infusion syndrome. Doses above 5mg/kg/hr should not be continued for more than 3-4 hours. In practice this means that the efficacy of propofol is limited, since safe doses may be insufficient to terminate electroencephalographic SE.
3. **Propofol monitoring:** In addition to triglycerides, creatine kinase and lactate monitoring, patients on continuous propofol infusions should have daily ECGs

and the dose should be reduced if there is a [Brugada pattern](#). A lipid profile should be obtained if a propofol infusion is required beyond three days.

8 Table 1: Continuous infusions of general anaesthesia for status epilepticus

Drug	Loading dose	Continuous Infusion Rate	Considerations
	Intravenous		
Midazolam	Dose: 0.03-0.3mg/kg Rate: 1mg over 30 seconds (eg 80kg patient, maximum of 24mg over 12 minutes)	Initial rate: 0.03-0.1mg/kg/hr Breakthrough: 0.1-0.2mg/kg bolus; increase infusion rate by 0.1mg/kg/hr every four hours until flat EEG or burst suppression. Maximum dose: 20mg/hr	Tachyphylaxis occurs; in long-term sedation with midazolam, larger doses may be required. When used with phenytoin, may displace phenytoin from albumin. Caution in interpreting phenytoin levels. See also IV monograph
Propofol	Dose: 1.0-2.5mg/kg Rate: 2ml every 10 seconds (using 2% solution) Consider lower end of range and slower rate of admin for patients >55yo.	Initial rate: 1-5mg/kg/hr (for short periods up to 12mg/kg/hr may be considered) Breakthrough: 1mg/kg bolus and increase infusion rate by 0.2mg/kg/hr every two hours until flat EEG or burst suppression. Maximum dose: 4mg/kg/hr	Periodically check: Triglycerides <5 Creatine Kinase <2000 Lactate <2.5 Monitor for: Propofol infusion syndrome (see maximum dose) Liaise with: Dietician to adjust lipid content of nutrition See also cautions and IV monograph
Ketamine	Induction: Usually 1mg/kg loading (range 0.4-4.5mg/kg)	Infusion: Initially 2mg/kg/hr (range 0.3-7.5mg/kg/hr)	Hypertension, potential neurotoxicity
Thiopental	Dose = 75mg-125mg over 10-15 seconds	Initial rate: 3mg/kg/hr Breakthrough: 1-2mg/kg bolus and increase infusion rate by 0.5mg/kg/hr every 6 hours Maximum dose: 800mg/hr	Very large volume of distribution (Vd); accumulates and may take many days to wear off. Risk of hyperkalaemia. See also cautions

8.1 Stage 3: Ongoing management (>60minutes) of refractory SE

Step 1: Start maintenance anticonvulsant during continuous infusion of general anaesthetics (GA)

First-line: **Levetiracetam IV/ NG/ PO 1000mg-1500mg twice daily**

Second-line: **Phenytoin IV 100mg three times daily** until stable levels, or adjust dose if levels out of range (first level within 48 hours)

Once stable convert to:

- NG/ PO if possible
- **Once** daily dosing, or **twice** daily if daily dose exceeds 400mg daily (divide dose)

Caution in women of child-bearing age

See also: CUH [Phenytoin in adults guideline](#)

Third-line: **Sodium Valproate IV – 10mg/kg daily**; divided in 2-3 doses

- Avoid in active liver disease, mitochondrial disease, porphyria
- Caution in women of child-bearing age
- Monitor for hyperammonaemia
- Loading dose if required = 20-30mg/kg at 6-10mg/kg/min.
- See also: national guidance on using [valproate in women and girls of childbearing years](#)

Step 2: Gradually (4-8hours taper) withdraw the continuous infusion GA after 24 hours under continuous EEG monitoring

- a. Consider additional phenobarbital bridging if first withdrawal of continuous infusion precipitated recurrent SE despite gradual taper. Patients with phenobarbital serum levels:
 - i. < 200mcg/ml are expected to breathe spontaneously,
 - ii. < 100mcg/ml are expected to be awake.

8.2 Recurrence of SE

If status epilepticus recurs despite a sufficiently long taper of continuously infused anaesthesia:

1. Repeat continuous infusion of general anaesthetic (see table 1) over 48 hours using an alternative agent
2. Add an additional anticonvulsant agent - Lacosamide 400mg IV/NG followed by 200mg IV/NG BD. Continuous ECG monitoring / medical supervision is required. Convert to oral / enteral as soon as able
3. Consider autoimmune encephalitis/immune-mediated RSE, and consider immunomodulatory treatment with pulse steroids (IV Methylprednisolone 1g per day for 3 days). Discuss indication for further immunomodulation with Neurology.
4. Consider hypothermia (32-35°C) for <48 hours
 - a. **Warning:** if using **propofol** consider an alternative agent before inducing hypothermia

5. Give 20mmol magnesium sulphate IV, and begin continuous infusion
 - a. Initially: 10mmol/hr (range 10-30mmol/hr)
 - b. Target serum level = 1.5mmol/l
6. Give pyridoxine 200mg/day (NG/PO) if adolescent (<18yo)

8.3 If continuing SE despite above measures

1. Reimaging
 - a. is there a catastrophic/ irreversible underlying brain injury?
2. Consider hypothermia (32-35°C) for <48 hours
 - a. **Warning:** if using **propofol** consider an alternative agent before inducing hypothermia
3. Consider ketogenic diet (4:1 ratio)
 - a. **Warning:** ketogenic diets should not be used in conjunction with **propofol**
 - b. Liaise with dietician
4. If lesional SE, consider neurosurgery
5. Consider vagal nerve stimulation (VNS), up to 1.25mA
6. Therapies of last resort:
 - a. CSF drainage
 - b. Electroconvulsive therapy (in liaison with a neurologist)

9 Typical treatment errors

- SE was never electroencephalographically controlled despite cessation of clinical convulsions (due to under-monitoring of EEG or inadequate burst suppression target) → monitor EEG
- SE was controlled, but inadequate or no maintenance anticonvulsant therapy initiated prior to withdrawal of continuous infusion → commence maintenance anticonvulsant therapy early
- SE was controlled, but withdrawal of continuous infusion too rapid → attempt a more gradual taper at next attempt

10 Treatment notes

- Multiple medications have been used to terminate RSE. The evidence base is weak for each, but treatment delay clearly reduces efficacy.
- There is no defined optimum duration of treatment or finite number of trials after which treatment is futile. Good recovery may occur after months of prolonged or intermittent SE. Continue treatment as long as no devastating brain injury is visible on MRI (the significance of high cortical signal on MRI is uncertain as this can be a reversible effect of seizure activity).

11 Monitoring compliance with and the effectiveness of this document

This document is a guideline and sets forward perceived best practice and is in line with the NICE clinical guidelines for epilepsy. However, it cannot be presumed to be the standard for all patients owing to the broad range of conditions that can result in seizures.

Patients who will benefit from the treatments present rarely and sporadically and cannot easily be audited. It is envisaged not to audit compliance with this guideline.

Patients will be managed by a combination of intensivists, neurologists, specialist nurses and pharmacists who will use this document as a resource.

12 References

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13 Associated documents

[Guidelines for GP: epilepsy guidelines.](#)
[Guideline: epilepsy in pregnancy and labour](#)
[Guideline: epileptic seizures during labour](#)
[Seizure observation chart](#)

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